

MPA vs SPA

MPA (Multi-Page Application)

- The server renders each page independently (full page load/navigation per URL).
- Best for document/content-heavy sites, strong SEO needs, and simple navigation flows.
- Pros: SEO and social preview friendliness out of the box, simpler progressive enhancement, predictable caching, smaller per-page payloads.
- Cons: More round-trips to server, page reloads break client state unless persisted, UX can feel less “app-like.”

SPA (Single-Page Application)

- The app loads a single HTML shell; navigation and UI updates happen client-side (via JS routing and DOM updates).
- Best for highly interactive, stateful experiences (dashboards, editors, realtime apps).
- Pros: Smooth app-like UX, rich client interactivity, fewer full page reloads, great for offline/desktop-like behavior.
- Cons: SEO/first-load challenges unless server-rendered (SSR), bigger initial bundle, complexity in caching and analytics, higher front-end engineering cost.

Key decision dimensions (what to evaluate)

When choosing, weigh these factors:

- **SEO & discoverability** — high → lean MPA or SSR/Hybrid.
 - **Interactivity & realtime updates** — high → lean SPA.
 - **Initial load / perceived performance** — critical → SSR/MPA or optimized SPA with code-splitting.
 - **Developer skills & maintenance** — small team → simpler MPA or framework with conventions.
 - **Offline / client state complexity** — high → SPA (or PWA).
 - **Third-party integration (analytics, bots, crawlers)** — prefer server rendering/hybrid.
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Recommended approach and why

E-commerce platform (catalog pages + cart + personalized UX)

Recommendation: Hybrid — SSR/MPA for catalog + SPA for cart/checkout (or SSR/Isomorphic SPA)

Why:

- Product/category pages need SEO and fast indexable content (favors server rendering or MPA).
- Cart, checkout, personalization, and account dashboards need smooth, stateful interactions (favors SPA behavior).
- A hybrid approach gives best of both worlds: SEO and fast first paint for product pages; app-like UX for transactional flows.

Trade-offs & notes:

- Two approaches:
 1. **MPA with client widgets:** server-render product pages; inject small SPA widgets for cart and personalization. Lower complexity, easier caching.
 2. **Isomorphic/SSR SPA:** one app that server-renders initial HTML and hydrates into a SPA (Next.js, Nuxt, Remix). Cleaner code sharing but more engineering overhead.
- Security and PCI considerations push checkout to tightly controlled flows; can be isolated as SPA micro-app or server-rendered secure pages.

Tech picks: Next.js / Remix with SSR and client routing; or traditional server pages + dedicated cart micro-frontend.

Quick comparison table (summary)

Project type	Recommended approach	Primary reason
Content / Marketing	MPA / SSG / SSR	SEO, caching, low interactivity
Internal Dashboard	SPA	High interactivity & realtime state
E-commerce	Hybrid (SSR + SPA parts)	SEO for catalog + smooth transactional UX